

ARE, Sweden

WMO: 022170

Lat: 63.3738N

Lon: 13.1607E

Elev: 403

StdP: 96.58

Time Zone: 1.00 (EUC)

Period: 01-16

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-24.6	-21.1	-26.9	0.3	-24.6	-23.2	0.5	-21.2	10.0	0.9	9.0	0.1	0.2	320	0.657

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	9.3	25.5	17.2	23.4	16.5	21.2	15.2	18.1	23.8	17.0	21.8	15.9	20.2	2.5	140

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
15.9	11.8	20.0	14.9	11.1	19.0	13.9	10.4	17.9	52.4	23.9	49.2	21.8	45.9	20.3	21.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 8.4	7.1	6.0	-27.7	28.3	3.4	1.6	-30.1	29.5	-32.1	30.4	-34.0	31.4	-36.4	32.6		
(5)			-27.6	19.3	3.0	1.1	-29.8	20.1	-31.5	20.8	-33.2	21.4	-35.4	22.2		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	3.2	-7.0	-5.7	-2.4	2.6	7.0	11.7	14.6	13.3	8.5	3.1	-2.0	-5.6		
	DBStd	8.76	6.80	6.16	4.77	3.39	3.54	3.73	3.27	3.06	3.17	3.84	4.65	6.62		
	HDD10.0	2836	527	441	384	223	108	23	2	6	64	214	358	484		
	HDD18.3	5527	785	674	643	473	351	201	124	159	294	471	608	743		
	CDD10.0	362	0	0	0	0	15	74	144	107	20	1	0	0		
	CDD18.3	11	0	0	0	0	0	2	8	2	0	0	0	0		
	CDH23.3	174	0	0	0	0	3	49	92	31	0	0	0	0		
	CDH26.7	18	0	0	0	0	0	2	15	1	0	0	0	0		
	WSAvg	2.4	2.3	2.5	2.7	2.5	2.7	2.6	2.2	1.9	2.2	2.4	2.2	2.2		
(15) Precipitation	PrecAvg	634	47	38	38	29	41	70	92	85	54	52	46	41		
	PrecMax	861	96	104	74	64	87	125	177	143	109	135	95	87		
	PrecMin	476	2	4	14	6	17	28	17	37	13	13	7	18		
	PrecStd	102	23	20	14	15	18	25	45	36	28	27	23	17		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	5.6	5.6	9.3	16.8	23.3	26.8	28.7	26.4	20.4	14.6	9.7	6.1		
		MCWB	3.2	3.1	4.3	10.0	15.2	17.0	18.9	17.9	13.7	10.1	7.3	3.5		
	2%	DB	3.9	4.0	6.2	13.2	19.9	24.9	26.1	23.9	17.5	12.0	7.5	4.7		
		MCWB	2.0	2.5	3.1	7.5	12.8	16.2	18.1	17.1	12.8	8.7	5.7	2.9		
	5%	DB	2.4	3.1	4.8	10.5	16.6	22.1	23.6	21.4	15.6	10.3	5.2	3.4		
		MCWB	1.2	1.7	2.7	5.9	11.0	15.1	17.0	15.7	11.7	7.7	3.7	2.0		
	10%	DB	1.3	1.8	3.6	8.3	13.7	19.0	21.5	19.2	13.7	8.7	3.6	2.4		
		MCWB	0.4	0.7	1.9	4.9	9.2	13.8	15.7	14.9	10.6	6.9	2.5	1.1		
	0.4%	WB	3.3	3.6	4.7	10.3	15.5	18.0	19.6	18.6	14.9	10.9	7.9	4.2		
		MCDB	5.0	4.6	7.8	16.7	22.4	24.9	26.7	24.3	18.6	13.8	9.9	5.3		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	2%	WB	2.3	2.8	3.8	7.8	13.2	16.7	18.6	17.5	13.4	9.1	5.8	3.1		
		MCDB	3.4	3.8	5.7	12.2	18.7	23.1	24.7	22.3	16.7	11.3	7.2	4.5		
	5%	WB	1.4	2.0	3.0	6.5	11.6	15.4	17.6	16.4	12.0	8.1	3.8	2.1		
		MCDB	2.3	2.9	4.6	9.9	16.1	21.5	22.7	20.4	15.2	10.1	4.9	3.2		
	10%	WB	0.3	0.6	2.1	5.3	9.7	14.0	16.3	15.4	10.8	7.0	2.6	1.3		
		MCDB	1.2	1.7	3.4	7.9	13.3	18.6	20.5	18.6	13.2	8.9	3.5	2.3		
(35) Mean Daily Temperature Range	5% DB	MDBR	6.2	6.7	8.2	8.5	8.8	9.2	9.3	8.7	7.5	5.8	5.1	6.2		
		MCDBR	5.8	6.5	8.8	13.1	14.7	15.2	13.8	12.7	10.8	8.8	6.9	6.3		
		MCWBR	4.9	5.3	6.2	7.7	8.4	7.3	6.6	6.4	6.5	5.8	5.4	5.2		
	5% WB	MCDBR	5.6	6.2	7.5	11.5	14.0	13.8	12.6	11.4	10.0	7.9	6.7	6.0		
		MCWBR	4.9	5.4	5.4	7.1	8.2	6.9	6.4	6.2	6.2	5.7	5.4	5.0		
(40) Clear-Sky Solar Irradiance	taub		0.199	0.260	0.291	0.322	0.332	0.330	0.341	0.330	0.303	0.285	0.209	N/A		
	taud		2.015	2.218	2.256	2.322	2.451	2.502	2.486	2.506	2.567	2.464	2.087	N/A		
	Ebn at Noon		435	686	809	860	884	893	870	850	811	668	414	N/A		
	Edh at Noon		29	58	85	99	96	93	93	83	64	47	26	N/A		
	RadAvg		0.17	0.70	2.06	3.90	4.61	4.89	4.71	3.57	2.12	0.91	0.26	0.08		
(45) All-Sky Solar Radiation	RadStd		0.02	0.10	0.23	0.31	0.33	0.44	0.54	0.35	0.20	0.10	0.03	0.01		

Historical Trends

		DBAvg	Heating			Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP		HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	
(47) Regional (46 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	-142	-178	N/A	N/A	

Nomenclature: See separate page